

NORTHEAST OHIO, OH, USA

WMO: 725256

Lat: 41.778N

Lon: 80.696W

Elev: 282

StdP: 97.99

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 04857

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -16.9	(c) -13.5	(d) -21.6	(e) 0.6	(f) -15.4	(g) -18.4	(h) 0.8	(i) -12.4	(j) 11.2	(k) -1.4	(l) 10.1	(m) -1.7	(n) 2.4	(o) 220	(p) 0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	31.4	23.1	30.0	22.3	28.6	21.6	24.5	29.2	23.6	27.9	22.8	26.8	4.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.3	26.9	22.3	17.5	26.2	21.4	16.6	25.2	75.3	29.5	71.7	27.7	68.5	26.7	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.5	(b) 8.4	(c) 7.5	(e) -21.2	(f) 33.4	(g) 4.9	(h) 1.2	(i) -24.7	(j) 34.2	(k) -27.6	(l) 34.9	(m) -30.4	(n) 35.6	(o) -33.9	(p) 36.5		
(a) 9.5	(b) 8.4	(c) 7.5	(e) -21.6	(f) 25.8	(g) 4.7	(h) 0.9	(i) -25.0	(j) 26.5	(k) -27.8	(l) 27.0	(m) -30.4	(n) 27.6	(o) -33.9	(p) 28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-2.9	-2.5	2.2	8.9	15.0	19.6	21.9	20.9	17.7	11.5	5.8	0.6
	DBStd	10.12	6.59	6.45	6.38	5.44	4.98	3.78	3.00	2.92	4.00	4.68	4.99	5.08
	HDD10.0	1586	403	353	256	84	12	0	0	0	1	37	146	294
	HDD18.3	3408	659	584	501	286	128	29	5	9	60	218	377	551
	CDD10.0	1567	2	2	13	52	167	288	369	339	231	84	18	2
	CDD18.3	348	0	0	1	3	25	67	116	90	41	6	0	0
	CDH23.3	2907	0	0	3	53	234	595	982	684	314	40	1	0
	CDH26.7	798	0	0	0	10	51	184	291	169	87	6	0	0
Wind	WSAvg	3.2	4.1	3.9	3.6	3.6	2.9	2.6	2.2	2.1	2.4	3.2	3.7	4.0
Precipitation	PrecAvg	1117	85	62	76	99	97	101	115	94	109	98	93	90
	PrecMax	1388	158	144	158	155	187	184	241	160	208	170	136	187
	PrecMin	756	39	23	23	59	26	20	36	48	30	49	43	49
	PrecStd	170	30	28	33	30	43	41	53	32	48	32	29	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	17.3	22.8	28.0	30.3	32.6	33.0	32.6	31.7	27.3	21.3	17.0
		MCWB	12.8	11.9	15.3	17.8	21.4	23.2	23.7	23.9	23.1	21.0	14.7	13.3
	2%	DB	12.6	12.5	18.4	24.6	28.0	31.0	31.3	30.2	29.1	23.9	17.8	13.0
		MCWB	10.5	9.8	12.8	15.5	19.8	22.7	23.3	22.8	22.3	18.0	13.1	10.2
	5%	DB	9.1	9.2	15.3	21.7	26.2	28.9	29.8	28.4	27.1	21.4	16.1	10.6
		MCWB	7.0	6.3	11.0	14.0	18.9	21.5	22.6	22.0	21.0	17.0	12.0	8.5
	10%	DB	6.2	6.2	12.4	18.6	23.8	27.0	28.2	27.2	24.7	19.1	13.7	7.8
		MCWB	4.0	3.4	8.7	12.5	17.2	20.2	21.6	21.2	19.7	15.5	10.2	5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.3	13.9	16.2	19.1	22.9	25.0	25.8	25.2	24.8	21.9	16.7	13.8
		MCDB	15.4	16.3	22.0	25.7	28.5	30.7	30.8	30.5	29.4	26.4	19.9	15.7
	2%	WB	10.7	10.0	13.9	16.8	21.1	23.6	24.5	24.1	23.2	19.8	14.1	11.0
		MCDB	12.4	12.1	17.6	23.0	26.3	28.7	29.3	28.2	27.0	22.5	16.7	12.7
	5%	WB	7.4	6.6	11.6	15.0	19.9	22.4	23.6	23.1	21.8	17.8	12.4	8.2
		MCDB	9.1	9.1	14.7	19.9	24.7	27.3	27.8	26.8	25.5	20.4	15.4	9.9
	10%	WB	4.3	3.5	8.8	13.2	18.6	21.3	22.7	22.2	20.6	15.8	10.5	5.8
		MCDB	6.2	6.0	12.3	18.0	22.7	25.5	26.6	25.7	23.8	18.8	13.3	7.8
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.7	9.3	11.1	11.7	11.1	10.8	10.7	10.8	9.4	7.9	6.3
		MCDBR	10.3	11.8	13.8	15.8	13.9	12.8	12.2	12.1	13.0	12.6	11.3	10.1
		MCWBR	9.1	9.7	9.1	9.2	7.5	6.6	5.8	6.3	6.8	7.4	8.1	8.6
	5% WB	MCDBR	10.0	11.5	12.4	14.0	12.1	11.1	10.1	10.0	11.3	10.8	10.1	9.5
		MCWBR	9.4	10.1	9.0	9.1	6.9	6.3	5.3	5.7	6.2	7.3	8.1	8.8
Clear-Sky Solar Irradiance	taub	0.302	0.334	0.349	0.383	0.419	0.443	0.443	0.428	0.384	0.362	0.319	0.300	
	taud	2.417	2.274	2.311	2.306	2.237	2.213	2.231	2.267	2.363	2.414	2.514	2.500	
	Ebn at Noon	847	863	895	886	858	836	831	833	849	825	825	822	
	Edh at Noon	79	106	115	125	137	141	137	128	109	92	72	68	
All-Sky Solar Radiation	RadAvg	1.41	2.19	3.26	4.45	5.36	5.92	5.93	5.25	4.19	2.65	1.68	1.06	
	RadStd	0.12	0.26	0.36	0.45	0.49	0.46	0.43	0.27	0.40	0.25	0.25	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+98	N/A

Nomenclature: See separate page